

SymPy Bug Card

Bug 8: k -weighted binomial sum drops the $n = 1$ exception

Task. Evaluate the k -weighted alternating binomial sum for a nonnegative integer n .

$$\sum_{k=0}^n (-1)^k k \binom{n}{k}$$

SymPy's answer.

$$\text{summation}\left((-1)^k k \binom{n}{k}, (k, 0, n)\right) = 0$$

Correct answer.

$$\begin{cases} -1, & n = 1, \\ 0, & n \in \mathbb{Z}_{\geq 0}, n \neq 1. \end{cases}$$

Explanation. At $n = 1$, the finite sum is $0 \binom{1}{0} - 1 \binom{1}{1} = -1$, but SymPy's result is the constant function 0 for all n .

Likely root cause. The diagnosis traces the result to the Gosper finite-sum branch in `sympy/concrete/gosper.py`: the certificate $-(k-1)/(n-1)$ has a pole at $n = 1$, but the endpoint formula simplifies to 0 before that excluded parameter is preserved. This is the same `gosper_sum` lost-endpoint failure family reported in the open issue [sympy/sympy#29731](#) (created 2026-05-02), which predates this bundle.

Artifact (GitHub).

[bug-report/bug-008-k-weighted-binomial-sum-drops-the-n-1-exception/](#)